

Mathematical Foundations of Political Methodology

Robert Gulotty

Department of Political Science
University of Chicago

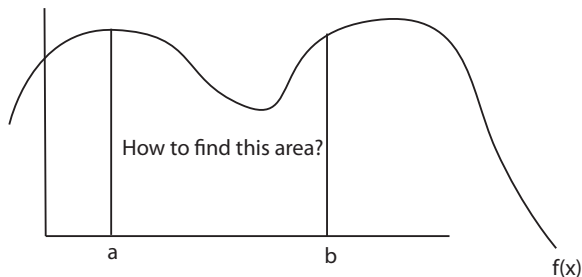
September 18, 2020

- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

Integral: Geometric Intuitions

“The opposite of differentiation,” “antiderivative,” or “taking an area”



Integral: Geometric Intuitions

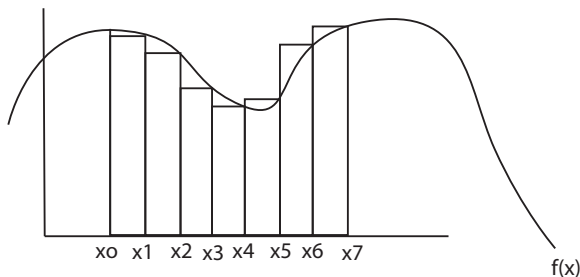
Let $a < b$ be real numbers and let f be a continuous function.

Integral: Geometric Intuitions

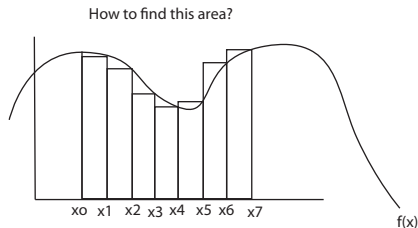
Let $a < b$ be real numbers and let f be a continuous function.

We want to find the area bounded by the curve $y = f(x)$, the vertical lines $x = a$ and $x = b$, and the x -axis. Let $a = x_0$ and $b = x_7$

How to find this area?

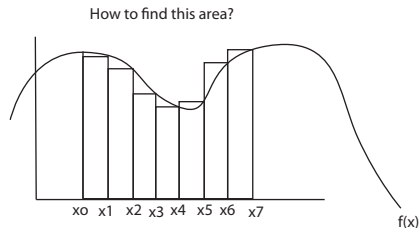


Integral: Geometric Intuitions



$$\sum_{i=1}^7 (x_i - x_{i-1}) f(x_i) \text{ or letting } \Delta x_i = x_i - x_{i-1} \text{ this approximation is}$$
$$\sum_{i=1}^7 f(x_i) \Delta x_i$$

Integral: Geometric Intuitions



$\sum_{i=1}^7 (x_i - x_{i-1})f(x_i)$ or letting $\Delta x_i = x_i - x_{i-1}$ this approximation is $\sum_{i=1}^7 f(x_i)\Delta x_i$

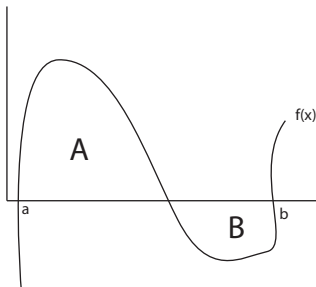
Letting $\lim_{\Delta x_i \rightarrow 0}$ gives us the solution; it's written

$$\int_a^b f(x)dx$$

Integral: Geometric Intuitions

Areas below the x -axis are negative:

$$\int_a^b f(x) = \text{Area}(A) - \text{Area}(B)$$



Integral: Definition

Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$. We will define the Riemann Integral as $\int_a^b f(x)dx$. If this exists then we say f is *integrable* on $[a, b]$ and call $\int_a^b f(x)dx$ the *integral* of f .

Features of an Integral

$$\int_a^b f(x)dx$$

- $\int$ is the indication for taking an integral.
- dx tells us that we are taking an integral with respect to x .
- a is the lower bound of the integration.
- b is the upper bound of the integration.

Example of Non-Integrable Function

Suppose $f : [0, 1] \rightarrow \frac{1}{x}$

$$\int_0^1 \frac{1}{x} dx$$

Example of Non-Integrable Function

Suppose $f : [0, 1] \rightarrow \frac{1}{x}$

$$\int_0^1 \frac{1}{x} dx$$

Then $\frac{1}{x}$ is not integrable on $[a, b]$ because the area that the integral would represent is infinite.

Example of Non-Integrable Function

Suppose $f : [0, 1] \rightarrow \frac{1}{x}$

$$\int_0^1 \frac{1}{x} dx$$

Then $\frac{1}{x}$ is not integrable on $[a, b]$ because the area that the integral would represent is infinite.

$$f(x) = \begin{cases} 1 & \text{if } x \text{ rational} \\ 0 & \text{if } x \text{ irrational} \end{cases}$$

Example of Non-Integrable Function

Suppose $f : [0, 1] \rightarrow \frac{1}{x}$

$$\int_0^1 \frac{1}{x} dx$$

Then $\frac{1}{x}$ is not integrable on $[a, b]$ because the area that the integral would represent is infinite.

$$f(x) = \begin{cases} 1 & \text{if } x \text{ rational} \\ 0 & \text{if } x \text{ irrational} \end{cases}$$

This function is called Dirichlet Function, it is not integrable, because every interval will contain a discontinuous jump

- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

Fundamental Theorem of Calculus

There is a connection between derivative and integral, the relationship is described by the Fundamental Theorem of Calculus.

Fundamental Theorem of Calculus

There is a connection between derivative and integral, the relationship is described by the Fundamental Theorem of Calculus. First, we introduce the antiderivative F :

Theorem

Given a number a and a continuous function $f(x)$ we can define another function F such that

$$F(t) = \int_a^t f(x) dx \text{ for all } t$$

Then F is differentiable and $F'(t) = f(t)$ for all t

Fundamental Theorem of Calculus

There is a connection between derivative and integral, the relationship is described by the Fundamental Theorem of Calculus. First, we introduce the antiderivative F :

Theorem

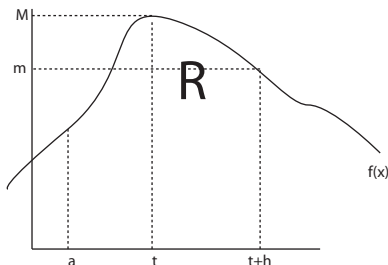
Given a number a and a continuous function $f(x)$ we can define another function F such that

$$F(t) = \int_a^t f(x) dx \text{ for all } t$$

Then F is differentiable and $F'(t) = f(t)$ for all t

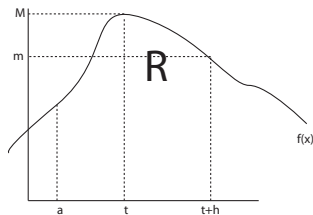
If you integrate a function and then differentiate the result, you retrieve the function that you started with $\Rightarrow$ “integration is the reverse of differentiation”

The Geometry of the Antiderivative F



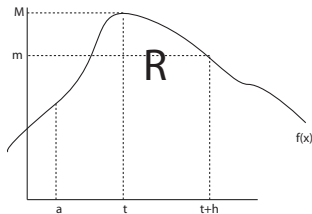
- $f(t+h) = m, f(t) = M$
- We will use mean value theorem to show that between t and $t+h$, there must be a box equal to the area under the curve.

The Geometry of the Antiderivative F



Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t .

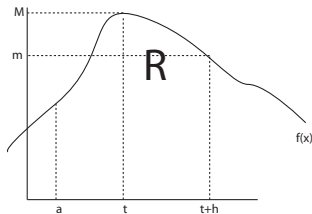
The Geometry of the Antiderivative F



Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t . Then

$$R = \int_a^{t+h} f(x)dx - \int_a^t f(x)dx = F(t+h) - F(t)$$

The Geometry of the Antiderivative F



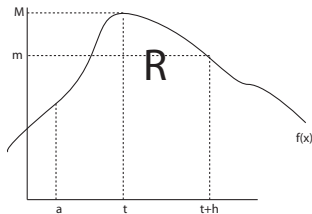
Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t . Then

$$R = \int_a^{t+h} f(x)dx - \int_a^t f(x)dx = F(t+h) - F(t)$$

The box made by $h \cdot m$ is smaller than R , the box made by $h \cdot M$ is bigger.

$$hm \leq F(t+h) - F(t) \leq hM$$

The Geometry of the Antiderivative F



Let $R = \int_t^{t+h} f(x)dx$ and let $F(t) = \int_a^t f(x)dx$ for all t . Then

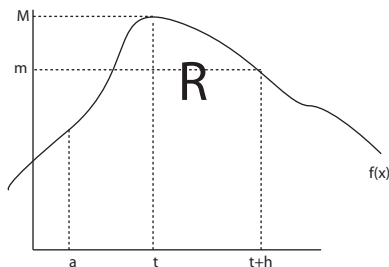
$$R = \int_a^{t+h} f(x)dx - \int_a^t f(x)dx = F(t+h) - F(t)$$

The box made by $h \cdot m$ is smaller than R , the box made by $h \cdot M$ is bigger.

$$hm \leq F(t+h) - F(t) \leq hM$$

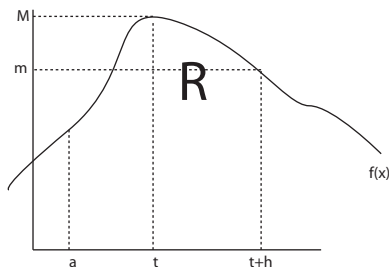
$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

The Geometry of the Antiderivative F



As $h \rightarrow 0$ we have $m \rightarrow f(t) = M$.

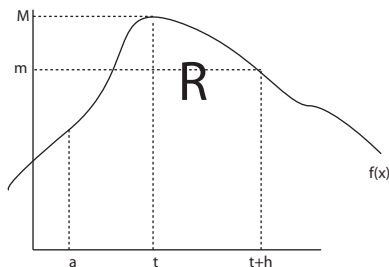
The Geometry of the Antiderivative F



As $h \rightarrow 0$ we have $m \rightarrow f(t) = M$. Thus,

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

The Geometry of the Antiderivative F



As $h \rightarrow 0$ we have $m \rightarrow f(t) = M$. Thus,

$$m \leq \frac{F(t+h) - F(t)}{h} \leq M$$

implies that

$$\lim_{h \rightarrow 0} \frac{F(t+h) - F(t)}{h} = f(t)$$

or that

$$F'(t) = f(t) \quad \square$$

Fundamental Theorem of Calculus

Theorem

Proposition: If F has a continuous derivative, then

$$\int_a^b f(x) = F(b) - F(a)$$

Fundamental Theorem of Calculus

Theorem

Proposition: If F has a continuous derivative, then

$$\int_a^b f(x) = F(b) - F(a)$$

$f(x)$ is the derivative of $F(x)$.

$F(x)$ is the antiderivative of $f(x)$.

Key idea, antiderivatives are not unique.

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.

By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.

By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.

$$f(x) = g'(x) = F'(x) \rightarrow g'(x) - F'(x) = 0$$

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.

By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.

$$f(x) = g'(x) = F'(x) \rightarrow g'(x) - F'(x) = 0$$

$$\text{So } (g(x) - F(x))' = 0$$

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.

By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.

$$f(x) = g'(x) = F'(x) \rightarrow g'(x) - F'(x) = 0$$

$$\text{So } (g(x) - F(x))' = 0$$

Recall, from the mean value theorem for $c < t < d$:

$$\frac{f(d) - f(c)}{d - c} = f'(t)$$

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.

By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.

$$f(x) = g'(x) = F'(x) \rightarrow g'(x) - F'(x) = 0$$

$$\text{So } (g(x) - F(x))' = 0$$

Recall, from the mean value theorem for $c < t < d$:

$$\frac{f(d) - f(c)}{d - c} = f'(t)$$

$$f(d) = f(c) + (d - c)f'(t)$$

Fundamental Theorem of Calculus

Start with $F' = f$ Define: $g(x) = \int_a^x f(t)dt$.

By the first bit, $g(x)$ is also an antiderivative of f on $[a, b]$.

$$f(x) = g'(x) = F'(x) \rightarrow g'(x) - F'(x) = 0$$

$$\text{So } (g(x) - F(x))' = 0$$

Recall, from the mean value theorem for $c < t < d$:

$$\frac{f(d) - f(c)}{d - c} = f'(t)$$

$$f(d) = f(c) + (d - c)f'(t)$$

if $f'(t)=0$, then

$$f(d) = f(c)$$

$f()$ is constant!.

Fundamental Theorem of Calculus

$(g(x) - F(x))' = 0 \rightarrow g(x) - F(x)$ is constant.

$$g(x) - F(x) = C$$

$$g(b) - F(b) = g(a) - F(a)$$

Fundamental Theorem of Calculus

$(g(x) - F(x))' = 0 \rightarrow g(x) - F(x)$ is constant.

$$g(x) - F(x) = C$$

$$g(b) - F(b) = g(a) - F(a)$$

$$g(b) = g(a) - F(a) + F(b)$$

Fundamental Theorem of Calculus

$(g(x) - F(x))' = 0 \rightarrow g(x) - F(x)$ is constant.

$$g(x) - F(x) = C$$

$$g(b) - F(b) = g(a) - F(a)$$

$$g(b) = g(a) - F(a) + F(b)$$

$$\int_a^b f(t)dt = \int_a^a f(t)dt - F(a) + F(b)$$

Fundamental Theorem of Calculus

$$(g(x) - F(x))' = 0 \rightarrow g(x) - F(x) \text{ is constant.}$$

$$g(x) - F(x) = C$$

$$g(b) - F(b) = g(a) - F(a)$$

$$g(b) = g(a) - F(a) + F(b)$$

$$\int_a^b f(t)dt = \int_a^a f(t)dt - F(a) + F(b)$$

$$\int_a^b f(t)dt = 0 - F(a) + F(b) \square$$

Apply Fundamental Theorem of Calculus

$$\int_a^b f'(x) dx = f(b) - f(a)$$

- Calculate **antiderivative**
- Evaluate at b
- Evaluate at a

Some Classic Antiderivative Formulas

antiderivative = indefinite integral

$$\int 1 dx = x + c$$

$$\int k dx = kx + c$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\int \frac{1}{x} dx = \log x + c$$

$$\int e^x dx = e^x + c$$

$$\int a^x dx = \frac{a^x}{\log a} + c$$

Example 1: Uniform Distribution

Suppose $f : \mathcal{R} \rightarrow \mathcal{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

Example 1: Uniform Distribution

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

What is the area under $f(x)$ from $[0, 1/2]$?

Example 1: Uniform Distribution

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

What is the area under $f(x)$ from $[0, 1/2]$?

$$\int_0^{1/2} f(x) dx = \int_0^{1/2} 1 dx$$

Example 1: Uniform Distribution

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

What is the area under $f(x)$ from $[0, 1/2]$?

$$\begin{aligned} \int_0^{1/2} f(x) dx &= \int_0^{1/2} 1 dx \\ &= x \Big|_0^{1/2} \end{aligned}$$

Example 1: Uniform Distribution

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

What is the area under $f(x)$ from $[0, 1/2]$?

$$\begin{aligned} \int_0^{1/2} f(x) dx &= \int_0^{1/2} 1 dx \\ &= x \Big|_0^{1/2} \\ &= (1/2) - (0) \\ &= 1/2 \end{aligned}$$

Example 1: Uniform Distribution

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = 1 \text{ if } x \in [0, 1]$$

$$f(x) = 0 \text{ otherwise}$$

What is the area under $f(x)$ from $[0, 1/2]$?

$$\begin{aligned}\int_0^{1/2} f(x) dx &= \int_0^{1/2} 1 dx \\ &= x \Big|_0^{1/2} \\ &= (1/2) - (0) \\ &= 1/2\end{aligned}$$

We will call $f(x) = 1$ the uniform distribution.

Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x)dx$

Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x)dx$

$$\int_2^t f(x)dx = \int_2^t xdx$$

Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x)dx$

$$\begin{aligned}\int_2^t f(x)dx &= \int_2^t xdx \\ &= \frac{x^2}{2} \Big|_2^t\end{aligned}$$

Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x)dx$

$$\begin{aligned}\int_2^t f(x)dx &= \int_2^t xdx \\ &= \left. \frac{x^2}{2} \right|_2^t \\ &= \frac{t^2}{2} - \frac{2^2}{2}\end{aligned}$$

Example 2: Area Under a Line

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$, with

$$f(x) = x$$

Evaluate the $\int_2^t f(x)dx$

$$\begin{aligned}\int_2^t f(x)dx &= \int_2^t xdx \\ &= \left. \frac{x^2}{2} \right|_2^t \\ &= \frac{t^2}{2} - \frac{2^2}{2} \\ &= \frac{t^2}{2} - \frac{4}{2} = \frac{t^2}{2} - 2\end{aligned}$$

- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

Integration Facts

Theorem

If $f_1, f_2 : [a, b] \rightarrow \mathbb{R}$ and f_1, f_2 are integrable on $[a, b]$, then

i) Consider the interval $[a, b]$ and $c \in [a, b]$. Then,

$$\begin{aligned}\int_c^c f_1'(x) dx &= f_1(c) - f_1(c) = 0 \\ \int_a^b f_1'(x) dx &= \int_a^c f_1'(x) dx + \int_c^b f_1'(x) dx \\ &= (f_1(c) - f_1(a)) + (f_1(b) - f_1(c)) \\ &= f_1(b) - f_1(a)\end{aligned}$$

Integration Facts

Theorem

If $f_1', f_2' : [a, b] \rightarrow \mathbb{R}$ and f_1', f_2' are integrable on $[a, b]$ and f_1' has antiderivative f_1 and f_2' has antiderivative f_2 , then

ii) For $c_1, c_2 \in \mathbb{R}$ then

$$\int_a^b (c_1 f_1'(x) + c_2 f_2'(x)) dx = c_1 \int_a^b f_1'(x) dx + c_2 \int_a^b f_2'(x) dx$$

- Integral
- Fundamental Theorem of Calculus
- Integration Facts
- Indefinite and Definite Integrals

Definite Integrals

With definite integrals we do not need to worry about the constant of integration:

Definite Integrals

With definite integrals we do not need to worry about the constant of integration: We know that

$$\int_a^b g'(x) dx = g(b) - g(a)$$

Definite Integrals

With definite integrals we do not need to worry about the constant of integration: We know that

$$\int_a^b g'(x) dx = g(b) - g(a)$$

If we used $g(x) + C$ as our primitive instead we'd get

Definite Integrals

With definite integrals we do not need to worry about the constant of integration: We know that

$$\int_a^b g'(x) dx = g(b) - g(a)$$

If we used $g(x) + C$ as our primitive instead we'd get

$$\int_a^b g'(x) dx = (g(b) + C) - (g(a) + C) = g(b) - g(a)$$

Integration Examples (Fearon 1998)

- The per period benefits of defecting are a , the cost of punishment is c , which occurs at time Δ . The payoff for cooperation is $\frac{z}{r}$. We continuously discount at rate r .
- The maximum benefit of a state for defecting from an agreement is:

$$\int_0^{\Delta} ae^{-rs} ds - \int_{\Delta}^{\infty} ce^{-rs} ds$$

Integration Examples (Fearon 1998)

- The per period benefits of defecting are a , the cost of punishment is c , which occurs at time Δ . The payoff for cooperation is $\frac{z}{r}$. We continuously discount at rate r .
- The maximum benefit of a state for defecting from an agreement is:

$$\begin{aligned} & \int_0^{\Delta} ae^{-rs} ds - \int_{\Delta}^{\infty} ce^{-rs} ds \\ &= \frac{1}{r} [a[-e^{-r*\Delta} + e^{-r*0}]] - \frac{1}{r} [ce^{-r*\infty} - -ce^{-r*\Delta}] \\ &= \frac{1}{r} [a[-e^{-r*\Delta} + 1] - ce^{-r*\Delta}] \end{aligned}$$

Integration Examples (Fearon 1998)

- The per period benefits of defecting are a , the cost of punishment is c , which occurs at time Δ . The payoff for cooperation is $\frac{z}{r}$. We continuously discount at rate r .
- The maximum benefit of a state for defecting from an agreement is:

$$\begin{aligned} & \int_0^{\Delta} ae^{-rs} ds - \int_{\Delta}^{\infty} ce^{-rs} ds \\ &= \frac{1}{r} [a[-e^{-r*\Delta} + e^{-r*0}]] - \frac{1}{r} [ce^{-r*\infty} - -ce^{-r*\Delta}] \\ &= \frac{1}{r} [a[-e^{-r*\Delta} + 1] - ce^{-r*\Delta}] \end{aligned}$$

- Cooperate if:

$$\frac{z}{r} \geq \frac{1}{r} [a[1 - e^{-r*\Delta}] - ce^{-r*\Delta}]$$

Integration Examples (Expectation)

- If X is a continuous random variable, it has a distribution $f(x)$ and satisfies the following:
 - For all $x \in \mathbb{R}$, $f(x) \geq 0$
 - $\int_{-\infty}^{\infty} f(x) dx = 1$
- The expected value of the random variable is $E[X] = \int_{-\infty}^{\infty} xf(x) dx$
- The uniform distribution over $[a, b]$ has pdf $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$ and $f(x) = 0$ otherwise